



KIRLOSKAR CHILLERS PRIVATE LIMITED

A Kirloskar Group Company

Enriching Lives

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Document Name: Recommended Water Quality Requirement for Chillers				
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Engineering Specification

The below data shall be referred for maintaining the standard water quality of circulation as well as make up water which is to be circulated from heat exchangers of chiller for longer and more economical life of the equipment.

Test Item		Cooling Water Line			Chilled water line		Tendency if the conditions are not met	
		Circulation type system		Once through type system	Circulation	Make up		
		Circulation	Make up				Corrosion	Scale
pH (@ 25°C)	-	6.5 ~ 8.2	6.0 ~ 8.0	6.8 ~ 8.0	6.8 ~ 8.0	6.8 ~ 8.0	Yes	Yes
Chloride Ion	ppm	below 200	below 50	below 50	below 50	Below 50	Yes	
Sulphuric Acid Ion	ppm	below 200	below 50	below 50	below 50	Below 50	Yes	
Total Hardness	ppm	below 200	below 70	below 70	below 70	Below 70		Yes
Calcium Hardness	ppm	below 150	below 50	below 50	below 50	below 50		Yes
Silica Ion	ppm	below 50	below 30	below 30	below 30	below 30		Yes
Iron	ppm	below 1.0	below 0.3	below 1.0	below 1.0	below 0.3	Yes	Yes
Copper	ppm	below 0.3	below 0.1	below 1.0	below 1.0	below 0.3	Yes	
Sulphide Ion	ppm	Should not be detected					Yes	
Ammonium Ion	ppm	below 1.0	below 0.1	below 1.0	below 1.0	below 0.1	Yes	
Residual Chlorine	ppm	below 0.3	below 0.3	below 0.3	below 0.3	below 0.3	Yes	
Free Carbon-di-oxide	ppm	below 4.0	below 4.0	below 4.0	below 4.0	below 4.0	Yes	
Ryzner Stability Index	RSI	6.0 ~ 7.0	---	---	---	---	Yes	Yes

Note: Kirloskar Chillers is not responsible for damage or under performance or faulty operation from untreated or improperly treated water.